Searching Short-Term Rentals in Porto and Lisbon

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Abstract

Choosing where to stay in a city is difficult when the available information is scattered: the accommodation description is biased because it is written by the host, the details that matter most about guests' experiences are buried in the reviews, and travellers can only see the surrounding points of interest if they scroll down the accommodation page. A search system that brings all this information together allows travelers to ask free-text questions, such as "quiet accommodations near historic sites", instead of filtering price and location alone. Such a system requires a broad range of data from various sources, combining general information about each accommodation, comprehensive reviews, a list of points of interest, and detailed descriptions of the most relevant tourist attractions. Combining these sources raises challenges: the data is in different formats, the reviews are multilingual and very unevenly distributed across listings, and points of interest have to be linked through their coordinates to the accommodations. We address them with a reproducible pipeline that collects, cleans and integrates the data from Inside Airbnb, OpenStreetMap and Wikipedia. The resulting collection is designed to answer the information needs that no single source covers on its own and provides a sound foundation for the indexing and retrieval stages that follow.

Keywords

information retrieval, short-term rentals, Airbnb, OpenStreetMap, Wikipedia, data preparation

1 Introduction

The goal of this project is the implementation of an information retrieval system. Our chosen topic is short-term rentals in Porto and Lisbon.

This paper covers the first milestone of that project: the preparation of the data it will search. It describes how the collection was obtained, processed and characterized.

Section 2.1 introduces the data sources, Inside Airbnb [1], OpenStreetMap [3] and Wikipedia [6], describing their origin, format, volume and license, and details how each one was collected: the Inside Airbnb downloads, the Overpass query used to extract points of interest (POIs), and the retrieval of Wikipedia articles through Wikidata [5]. Section 2.2 describes the pipeline that turns the raw data into the final collection, detailing each cleaning and integration step. Section 2.3 presents the conceptual data model. Section 2.4 characterizes the collection with descriptive and exploratory statistics, and Section 2.5 defines the prospective search tasks: information needs about nearby places, accommodation features and review content.

2 Milestone 1 – Data Preparation

2.1 Data Collection

One of our main concerns when choosing the topic for this project was to find high-quality data sources that contemplated useful data for our final purpose. We chose both Porto and Lisbon because they are the two largest cities in Portugal and its main tourist destinations, and to have a higher volume of data to feed the search system. The final choice was the datasets made available for download by Inside Airbnb, which contain all the important attributes of the listings and free-text reviews. In order to add a layer of information about the surroundings of each listing, we also considered OpenStreetMap, namely the Overpass API [4], to retrieve POIs such as museums, restaurants and transportation hubs. Finally, we also considered Wikipedia, which provides detailed descriptions of the most relevant attractions. All raw data is downloaded by a single `make collect` command, which takes about 20 minutes. Interrupted runs can be resumed without corrupting earlier downloads. Table 1 summarizes the raw data obtained from each source.

Inside Airbnb. Inside Airbnb is an independent project that scrapes the public Airbnb pages and makes the data available under

a CC BY 4.0 license. It is widely used in research and policy studies on short-term rentals, and it is not affiliated with Airbnb. We use the snapshot of 23 June 2026 for Porto and Lisbon since it is the most recent at the time of the data collection, and download three files per city: listings, reviews and neighbourhoods.

OpenStreetMap. OpenStreetMap is a collaborative project to create a free editable map of the world. Data coverage is detailed in urban areas but uneven across features: for instance, only a small fraction of POIs link to Wikidata. We extract POIs from it through the Overpass API, with one query per city restricted to a bounding box covering its metropolitan area, as Inside Airbnb contains listings for the whole Porto and Lisbon metropolitan areas. The query includes six categories relevant to a traveller: food and drink, tourism and heritage, transport, shops, leisure and public parking.

Wikipedia. Wikipedia is a free online encyclopedia that anyone can edit, but with community reviewing. Wikipedia articles are linked to POIs through their `wikidata` tag, present in 1,196 POIs. The Wikidata identifiers are resolved to their Wikipedia page. We then retrieve the plain text of each article with the TextExtracts API [2], using a script. Because of the geographic area considered, articles written in Portuguese are preferred, with English as a fallback.

File	Format	Porto	Lisbon	Size (MB)
Inside Airbnb [1] (CC BY 4.0)				
Listings	CSV	15,278	24,876	23.7
Reviews	CSV	1,131,143	1,886,932	395.2
Neighbourhoods	GeoJSON	177	173	7.2
OpenStreetMap [3] (ODbL 1.0)				
Points of interest	JSON	38,390	45,162	26.2
Wikipedia [6] (CC BY-SA 4.0)				
Articles*	JSON Lines	311	472	3.2

*782 distinct articles (766 Portuguese, 16 English); one is linked to both cities.

Table 1: Raw data per source.

2.2 Data Processing

2.2.1 *Pipeline Design*

The pipeline that turns the raw data into the final collection is shown in Figure 3. It has two stages, both driven by a Makefile: `make collect` downloads the raw files described in Section 2.1, and `make process` runs a single Python script that cleans each source and integrates them into one collection.

The four sources are first cleaned independently (Section 2.2.2). This keeps 30,590 of the 40,154 listings, 2,403,239 of the 3,018,075 reviews, the 49,615 named POIs out of 83,552 OSM elements, and all 782 Wikipedia articles, without their final sections. The cleaned sources are then integrated (Section 2.2.3): articles are attached to their POIs, each listing is linked to the POIs within 500 m, and each listing becomes one document, with its reviews and nearby POIs nested in it.

The pipeline is reproducible: the Inside Airbnb snapshot date is fixed, the Overpass query is versioned with the code, and all thresholds are defined as constants in the processing script. The cleaned intermediate tables are saved to `data/processed`, so each step can be inspected on its own, and `make analyze` recomputes every statistic in Section 2.4. Running `make` rebuilds the whole collection, and the processing stage reruns whenever the script or any raw file changes.

2.2.2 *Data Cleaning*

The raw data from Inside Airbnb requires significant cleaning to be useful for an information retrieval system. We applied a series of filtering and transformation rules to both the listings and their respective reviews to ensure the quality and relevance of the final collection.

First, we reduced noise by dropping features that do not contribute to the search process or are entirely empty. Columns with a 100% missing value rate, such as `neighbourhood`, `host_since`, and `calendar_updated`, were completely removed to free up memory and streamline the dataset.

Second, we transformed data types to enable quantitative filtering. The `price` column, originally formatted as a string containing currency symbols and commas (e.g., "$100.00"), was converted to a numerical float using regular expressions.

Third, we ensured the collection only contains active and searchable accommodations. Since our prospective search tasks (Section 2.5) rely heavily on textual evidence from previous guests, we removed listings with zero reviews. Furthermore, to guarantee the currency of the accommodations, we filtered out inactive listings by requiring at least one review within the last three years, using the `last_review` feature as a proxy for recent activity. Listings missing critical fields necessary for indexing and geographic cross-referencing—namely `name`, `description`, `latitude`, `longitude`, and `price`—were also discarded.

Finally, the review dataset was aggressively cleaned to maximize its informational value. We enforced referential integrity by dropping any reviews associated with the discarded listings. We then removed reviews with empty comments and established a minimum length threshold of 50 characters. This crucial step eliminates short, uninformative feedback (such as "ok" or "nice place") and ensures the remaining textual content has sufficient vocabulary density to accurately match complex user queries.

2.2.3 *Integration and Document Generation*

Integration links the cleaned sources around the listing, following the conceptual model in Section 2.3. Wikipedia articles are attached to POIs first, through the OSM element identifier recorded when each article was retrieved from a POI's `wikidata` tag, not by matching names. 778 of the 782 articles describe at least one of the 49,615 named POIs, giving 828 POIs with an article, since several OSM elements can refer to the same place.

Listings and POIs share no identifier, so they are linked by their coordinates: both are converted to Cartesian coordinates in metres and indexed in k-d trees, which return every listing–POI pair within 500 m. As central listings have many POIs in this radius, only the five nearest of each category are kept, together with every POI that has a Wikipedia article, as these are the landmarks most likely to be named in a query. The result is the `Proximity` association: 897,237

listing–POI pairs, each with its distance, linking 30,012 listings to 24,908 distinct POIs.

Finally, one JSON document per listing is written to `documents.jsonl`, with its reviews and nearby POIs as nested child documents, which map directly onto the nested documents of Solr. Each child has its own identifier and a `doc_type` field, so reviews and POIs can be searched on their own while remaining tied to their listing. POI children store their distance to the listing and, when available, the title, URL and text of their article, so that a query about a landmark can reach the listings around it. An abridged example is shown in Figure 4, in the appendix.

The collection occupies 2.7 GB, mostly review text and Wikipedia articles, which are copied into every listing near the POI they describe so that each document is self-contained. 360 listings end up without reviews, as all of theirs were shorter than 50 characters, and 578 have no POI within 500 m. Both are kept, since they can still be found through their description and amenities.

2.3 Conceptual Data Model

The conceptual data model, shown in Figure 1, consists of seven entities and is centred on the `Listing`, which is managed by a `Host`, is located in a `Neighbourhood` of a `City`, and receives `Reviews` from guests.

Each listing is also linked to the POIs in its surroundings through the `Proximity` association class, which stores the distance to each POI. Some POIs are described by a `WikipediaArticle`.

In the final collection, each document is a listing that contains its reviews and nearby POIs.

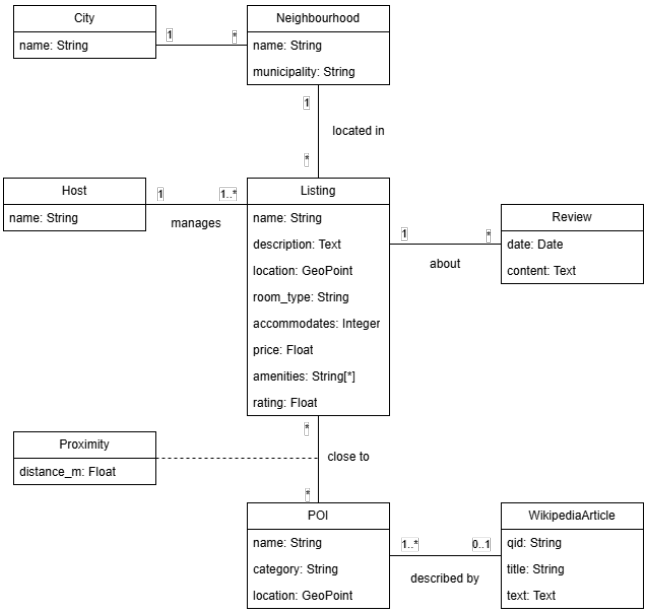


Figure 1: Conceptual data model.

2.4 Data Exploration

2.4.1 *Collection Overview*

The final collection has 30,590 documents, one per listing: 18,872 in Lisbon and 11,718 in Porto. Together they contain 2,403,239 reviews and 24,908 distinct POIs, of which 828 carry one of the 778 Wikipedia articles. The processing kept 76% of the listings and 80% of the reviews.

2.4.2 *Data Quality*

Before processing, we checked the raw files for the problems that would affect indexing. In the Inside Airbnb listings, 12 of the 90 columns are empty. The price is stored as text with a currency symbol (e.g. “$100.00”), and it is missing for 3,832 listings (9.5%). The amenities are a JSON list stored inside a CSV cell. There are no duplicated identifiers, and all coordinates are present and valid.

The review text fields contain HTML tags or entities, mostly line breaks. 4,712 listings have no reviews and 1,675 had their last review more than three years before the snapshot. Of all reviews, 209 have no text and 367,062 (12%) have fewer than 50 characters.

In OpenStreetMap, 33,937 of the 83,552 elements have no name, and the 35,805 ways and relations have no coordinates of their own, only the centre returned by Overpass. Only 1,196 elements have a `wikidata` tag. Of the 782 Wikipedia articles, 745 end with sections such as references, external links, see also or bibliography, which are not useful for search, and 122 have fewer than 100 words, mostly short entries for metro stations. These observations drove the cleaning rules in Section 2.2.2.

2.4.3 *Descriptive Statistics*

Entire homes make up 81% of the listings and private rooms 18%; hotel and shared rooms are rare. Prices are strongly right-skewed: the median is €136 in Lisbon and €117 in Porto, the 95th percentile is €407 and €323, and 167 listings cost more than €1,000. Entire homes have a median price of €156 in Lisbon and €124 in Porto, about twice that of private rooms. Ratings are concentrated at the top of the scale: the median is 4.77 in Lisbon and 4.81 in Porto, only 3% of the listings are rated below 4, and almost half are rated 4.8 or more, so the rating alone hardly distinguishes listings.

Reviews are unevenly distributed. A listing has a median of 34 reviews, but the 95th percentile is 313 and the maximum is 1,272, and the 10% most reviewed listings hold 44% of all reviews. Listings are also concentrated in the historic centres: in Porto, 51% of them are in a single parish union (Cedofeita, Santo Ildefonso, Sé, Miragaia, São Nicolau e Vitória), while in Lisbon the most common parish, Santa Maria Maior, has 16%.

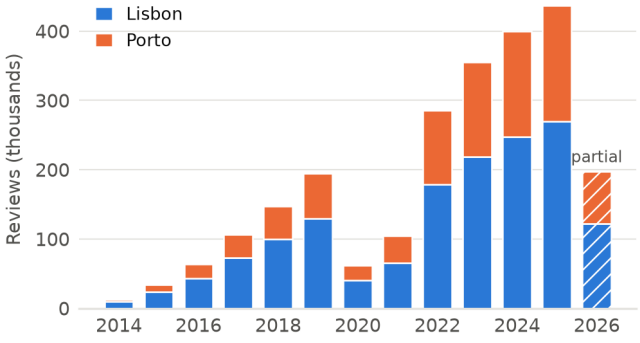


Figure 2: Reviews per year in the final collection. 2026 only covers the reviews until the snapshot.

The number of reviews per year, shown in Figure 2, grew until 2019, fell to 61,860 in 2020 with the COVID-19 pandemic, and has risen every year since, reaching 436,521 in 2025. Earlier years are under-represented, since only the reviews of listings active in the last three years were kept.

A listing has a median of 28 POIs within 500 m. The nearest food and drink place is at a median of 43 m, the nearest transport stop at 94 m and the nearest tourist or heritage POI at 102 m. Leisure and parking are less common nearby: 26% of the listings have no park or beach within 500 m. 80% of the listings have at least one POI with a Wikipedia article in that radius, with a median of five.

2.4.4 *Relationships Between Variables*

We measured the association between variables with Spearman's coefficient. Price is mainly related to size: its correlation with the number of guests is 0.70 in Lisbon and 0.61 in Porto, and with the number of bedrooms 0.60 and 0.59. It is almost unrelated to the number of reviews and only weakly related to the rating (0.23 and 0.18).

Location explains other patterns. The number of nearby POIs and the number of nearby POIs with a Wikipedia article are almost perfectly correlated (0.97 and 0.95), because both are concentrated in the historic centres. Listings with more POIs around them also have more reviews (0.29 and 0.26) and fewer bedrooms: central listings are small apartments with a high turnover, while larger houses are found further out.

2.4.5 *Text Analysis*

Most of the searchable text is in the reviews, which total 116 million words. A review has a median of 37 words and the 95th percentile is 129. Listing descriptions have a median of 67 words, with little variation, since the platform limits their length. The reviews are written in 85 languages: 57% are in English, 16% in French, 8% in Portuguese, 8% in Spanish and 6% in German, with French and Spanish more frequent in Porto. Wikipedia articles, on the other hand, are almost all in Portuguese, with a median of 315 words after cleaning and a long tail of up to 9,084 words. A query in English about a landmark will therefore often have to match a Portuguese article, which will have to be considered in the indexing stage.

The amenities are a separate source of text: a listing has a median of 35, and there are 4,902 distinct amenity strings, since hosts can write their own variants (e.g. "Crib - available upon request"). Finally, we looked at how often the terms used in our search tasks appear in the reviews: "quiet" appears in 5.7% of them, "noise" or "noisy" in 3.6%, "metro" in 5.2% and "check-in" in 7.4%, while "elevator" or "lift" appears in only 1.3% and "dog" in 0.3%. Information needs such as IN5 and IN7 will therefore depend on a small fraction of the reviews.

2.5 Prospective Search Tasks

The collection is designed to support ad hoc search: a traveller describes what they are looking for in free text, and the system returns a ranked list of accommodations.

- **IN1 – Quiet stay near historic sites.** *Query:* "quiet apartment near historic monuments in Porto". *Relevant:* a listing in Porto within walking distance of a historic POI, such as the Sé do Porto, whose description or reviews describe it as quiet.
- **IN2 – Family beach holiday.** *Query:* "family apartment close to the beach in Lisbon". *Relevant:* a listing in Lisbon near a beach POI, such as Praia de Carcavelos, that accommodates at least four guests and whose reviews mention families or children.
- **IN3 – Late arrival.** *Query:* "easy self check-in for a late arrival". *Relevant:* a listing whose reviews mention a smooth check-in late at night or a host who responds quickly. This need almost entirely relies on the review text.
- **IN4 – Travelling without a car.** *Query:* "apartment near the metro, no car needed". *Relevant:* a listing within a short walk of a metro or train station whose reviews mention good access to public transport.
- **IN5 – Accessible stay.** *Query:* "apartment with a functioning elevator". *Relevant:* a listing whose description or amenities mention an elevator, and whose reviews do not report it as broken or out of service. A listing with an elevator that does not work is not relevant.
- **IN6 – Nightlife.** *Query:* "apartment close to bars and nightlife in Lisbon". *Relevant:* a listing in Lisbon with several bars within about 300m, whose reviews mention nightlife or a lively area. Listings advertised as quiet and isolated are not relevant.
- **IN7 – Travelling with a dog.** *Query:* "pet friendly apartment near a park for walking the dog". *Relevant:* a listing whose amenities or description allow pets and that is within about 500m of a park POI, and whose reviews mention dogs or pets. Listings that forbid pets are not relevant.

3 Conclusions and Future Work

This milestone produced a collection of 30,590 listing documents that links Inside Airbnb, OpenStreetMap and Wikipedia through a reproducible pipeline. Its characterization showed that most searchable text is in multilingual, unevenly distributed reviews, while landmark descriptions are mostly in Portuguese. The next milestone will index the collection in Solr and evaluate the information needs defined in Section 2.5.

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A Appendices

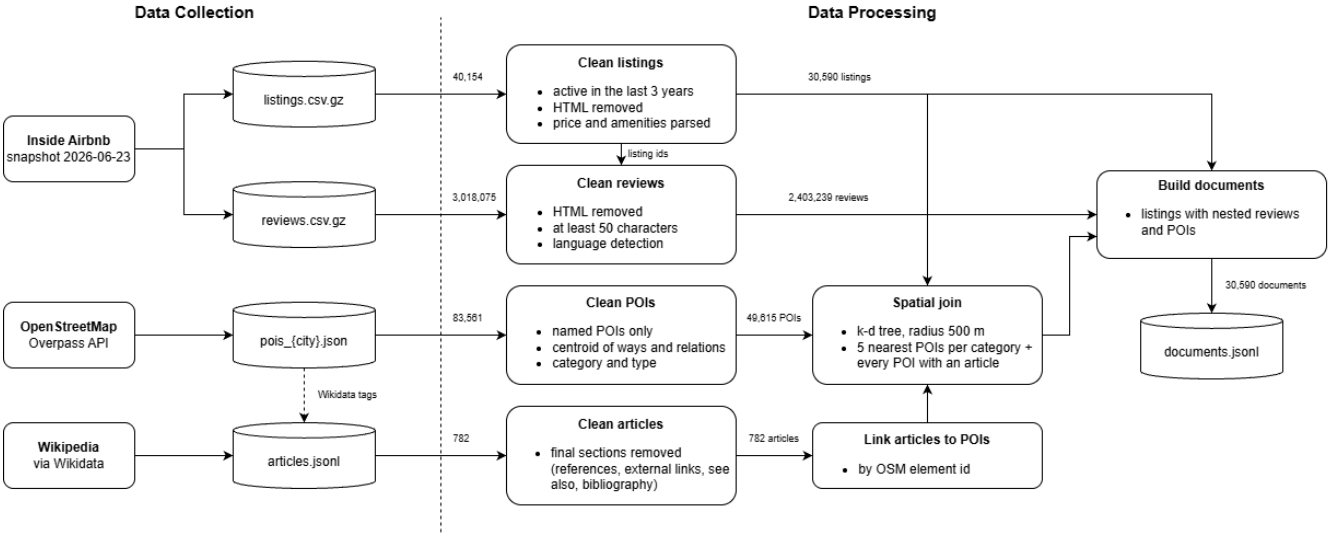


Figure 3: Data collection and processing pipeline.


```
{
  "id": "listing_36664931",
  "doc_type": "listing",
  "listing_id": 36664931,
  "listing_url": "https://www.airbnb.com/rooms/36664931",
  "name": "Home @ Porto Clérigos Apartment 4 – Studio",
  "description": "The Home @ Accommodations – Porto Clérigos Apartments are
    designed for people who travel to Porto and want to feel like they are at
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  "latitude": 41.15794372558594,
  "longitude": -8.629104614257812,
  "property_type": "Entire serviced apartment",
  "room_type": "Entire home/apt",
  "accommodates": 2,
  "price": 108.5,
  "amenities": ["Wifi", "Kitchen", "Self check-in", "Bathtub", "Crib –
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  "review_scores_rating": 4.75,
  "last_review": "2025-02-06",
  "reviews": [
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      "doc_type": "review",
      "date": "2025-02-06",
      "reviewer_id": 17644191,
      "reviewer_name": "Deborah",
      "comments": "Hugo's place is very well located to many local attractions.
    Hugo arranged car service to and from airport. [...]",
      "lang": "en"
    }
  ],
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      "osm_id": "node/9300234980",
      "name": "Café Léguas",
      "category": "food and drink",
      "type": "cafe",
      "distance_m": 125.5
    },
    {
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      "doc_type": "poi",
      "osm_id": "way/603359226",
      "name": "Casa da Música",
      "category": "tourism and heritage",
      "type": "attraction",
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      "wikipedia_title": "Casa da Música",
      "wikipedia_url": "https://pt.wikipedia.org/wiki/Casa_da_Música",
      "wikipedia_text": "A Casa da Música é uma sala de concertos e a segunda
    maior da cidade, localizada na Avenida da Boavista, no Porto [...]"
    }
  ]
}
```


Figure 4: Example of a document in the final collection.